Product info

KonstruX ST ZK 6,5x140		**Number of screws per joint 2**	
Article number 904811	100 Piece	Diameter screw / thread	6.5 mm
Length	140 mm	Diameter head	8.0 mm
Length screw head	5.5 mm	Diameter core	4.5 mm

User input

Side flaps

Pinewood | C24 | Spruce, pine, fir | Not pre-drilled
Width = 100 mm | Height = 120 mm | Overlap = 250 mm

Carrier

Pinewood | C24 | Spruce, pine, fir | Not pre-drilled
Width = 160 mm | Height = 240 mm

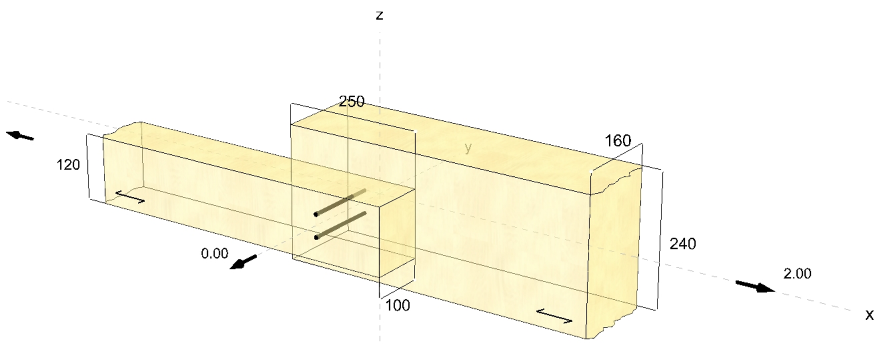
Load

Class of usage: 1

Design loads
x-direction = 2.00 kN | Duration of load: medium

Screws

Configuration: straight | Screwing: flush beam



Screw distances

Screwing angle	90 °
Drilling depth	0 mm

Distances - Side flaps [mm]	Minimum	Existing	
$a_{3,c}$	98	125	EN 1995-1-1
$a_{4,t}$	33	40	EN 1995-1-1
a_2	33	40	EN 1995-1-1

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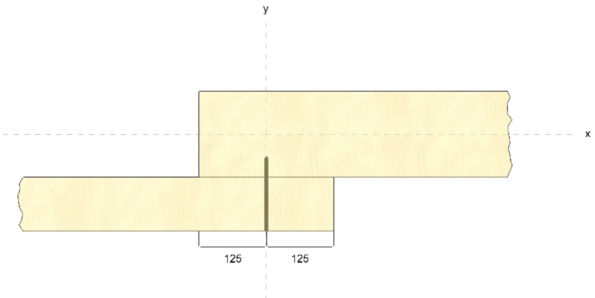

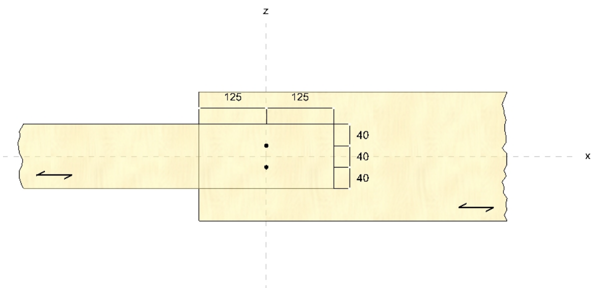

Design

Design loads

$V_{d,S} = F_{v,d} = 2.00\,kN$

Stress in x-direction

Thread in the side flaps

$n = 1$

$n_{ef} = n^{0.9} = 1.00$ — EN 1995-1-1 8.7.2 (8) (8.41)

$\alpha = 90\,°$ — ETA-11/0024

$k_{ax} = 1.0$ — ETA-11/0024

$f_{ax,k} = 11.40\,\frac{N}{mm^2}$ — ETA-11/0024

Screw calculated as bolt?

$d = 6.5\,mm$ — ETA-11/0024

$l_{ef} = 95\,mm$

$\rho_k = 350\,\frac{kg}{m^3}$ — EN 338 5, EN 14080 5.1.4.3 (4)(5)

$F_{ax,\alpha,Rk} = n_{ef} \cdot k_{ax} \cdot f_{ax,k} \cdot d \cdot l_{ef} \cdot \left(\frac{\rho_k}{350}\right)^{0,8} = 7.00\,kN$ — ETA-11/0024

Thread in the support

$n = 1$

$n_{ef} = n^{0.9} = 1.00$ — EN 1995-1-1 8.7.2 (8) (8.41)

$\alpha = 90\,°$ — ETA-11/0024

$k_{ax} = 1.0$ — ETA-11/0024

$f_{ax,k} = 11.40\,\frac{N}{mm^2}$ — ETA-11/0024

$d = 6.5\,mm$ — ETA-11/0024

$l_{ef} = 36\,mm$

$\rho_k = 350\,\frac{kg}{m^3}$ — EN 338 5, EN 14080 5.1.4.3 (4)(5)

$F_{ax,\alpha,Rk} = n_{ef} \cdot k_{ax} \cdot f_{ax,k} \cdot d \cdot l_{ef} \cdot \left(\frac{\rho_k}{350}\right)^{0,8} = 2.65\,kN$ — ETA-11/0024

Tensile load capacity

$n = 1$

$n_{ef} = n^{0.9} = 1.00$ — EN 1995-1-1 8.7.2 (8) (8.41)

$f_{tens,k} = 17.00\,kN$ — ETA-11/0024

$F_{t,Rk} = n_{ef} \cdot f_{tens,k} = 17.00\,kN$ — EN 1995-1-1 8.7.2 (7) (8.40c)

Shear

$V_{d,S} = 2.00\,kN$

$k_{mod,1} = 0.80$ EN 1995-1-1 3.1.3 (1)

$k_{mod,2} = 0.80$ EN 1995-1-1 3.1.3 (1)

$k_{mod} = \sqrt{k_{mod,1}k_{mod,2}} = 0.80$ EN 1995-1-1 2.3.3.1 (2) (2.6)

$n_{0,1} = 1$

$n_{ef,0,1} = 1.00$ EN 1995-1-1 8.3.1.1 (8) (8.17)

$n_{90,1} = 2$

$n_{0,2} = 1$

$n_{ef,0,2} = 1.00$ EN 1995-1-1 8.3.1.1 (8) (8.17)

$n_{90,2} = 2$

$n = min(n_{ef,0,1}\cdot n_{90,1}\,;n_{ef,0,2}\cdot n_{90,2}) = 2.00$

$\rho_{k,1} = 350\,\frac{kg}{m^3}$ EN 338 5, EN 14080 5.1.4.3 (4)(5)

$\rho_{k,2} = 350\,\frac{kg}{m^3}$ EN 338 5, EN 14080 5.1.4.3 (4)(5)

$f_{h,1,k} = 16.37\,\frac{N}{mm^2}$ ETA-11/0024

$f_{h,2,k} = 16.37\,\frac{N}{mm^2}$ ETA-11/0024

$t_1 = 100\,mm$ EN 1995-1-1 8.2.2 (1)

$t_2 = 40\,mm$ EN 1995-1-1 8.2.2 (1)

$\beta = \frac{f_{h,2,k}}{f_{h,1,k}} = 1.00$ EN 1995-1-1 8.2.2 (1) (8.8)

$M_{y,k} = 15.0\,Nm$ ETA-11/0024

$F_{ax,Rk} = 2.65\,kN$ EN 1995-1-1 8.2.2 (2)

(a) $f_{h,1,k}t_1 d = 10.64\,kN$ EN 1995-1-1 8.2.2 (1) (8.6)

(b) $f_{h,2,k}t_2 d = 4.26\,kN$ EN 1995-1-1 8.2.2 (1) (8.6)

(c) $\frac{f_{h,1,k}t_1 d}{1+\beta}\left[\sqrt{\beta+2\beta^2\left[1+\frac{t_2}{t_1}+\left(\frac{t_2}{t_1}\right)^2\right]+\beta^3\left(\frac{t_2}{t_1}\right)^2}-\beta\left(1+\frac{t_2}{t_1}\right)\right]+\frac{F_{ax,Rk}}{4} = 4.22\,kN$ EN 1995-1-1 8.2.2 (1) (8.6)

(d) $1{,}05\,\frac{f_{h,1,k}t_1 d}{2+\beta}\left[\sqrt{2\beta(1+\beta)+\frac{4\beta(2+\beta)M_{y,k}}{f_{h,1,k}dt_1^2}}-\beta\right]+\frac{F_{ax,Rk}}{4} = 4.54\,kN$ EN 1995-1-1 8.2.2 (1) (8.6)

(e) $1{,}05\,\frac{f_{h,1,k}t_2 d}{1+2\beta}\left[\sqrt{2\beta^2(1+\beta)+\frac{4\beta(1+2\beta)M_{y,k}}{f_{h,1,k}dt_2^2}}-\beta\right]+\frac{F_{ax,Rk}}{4} = 2.52\,kN$ EN 1995-1-1 8.2.2 (1) (8.6)

(f) $1{,}15\sqrt{\frac{2\beta}{1+\beta}}\sqrt{2M_{y,k}f_{h,1,k}d}+\frac{F_{ax,Rk}}{4} = 2.72\,kN$ EN 1995-1-1 8.2.2 (1) (8.6)

$F_{v,Rk} = min(F_{v,Rk}(a);F_{v,Rk}(b);F_{v,Rk}(c);F_{v,Rk}(d);F_{v,Rk}(e);F_{v,Rk}(f)) = 2.52\,kN$ EN 1995-1-1 8.2.2 (1) (8.6)

$\gamma_M = 1.30$ ÖNORM B 1995-1-1 2.4.1(1) P

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$$F_{v,Rd} = k_{mod} \cdot \frac{F_{v,Rk}}{\gamma_M} = 1.55\,kN$$

EN 1995-1-1
2.4.3 (1)P (2.17)

$$\eta = \left(\frac{V_{d,S}}{n \cdot F_{v,Rd}} \right) \cdot 100\,\% = 64.42\,\%$$

The system has been calculated successfully

Hints

- The calculation is according to:
- EN 338 (2016-07) + EN 14081-1 (2016-06) EN 14080 (2013-09)
- EN 1990 (2010-12)
- EN 1991-1-1 (2010-12)
- EN 1993-1-1 (2010-12)
- EN 1995-1-1 (2010-12) + EN 1995-1-1/A2 (2014-07) + ÖNORM B 1995-1-1 (2015-06)
- DIN EN 1995-1-1/NA (2013-08)
- ETA-11/0024 (2017-03)
- The screws may only be used when subjected predominantly to dead loads.
- Verification of the components must be carried out separately if necessary.
- Attention: Please check the assumptions. The displayed values, types and numbers of fasteners are dimensioning aids only.
  Projects can only be dimensioned by authorized people.
  If you need a verification of stability please contact an authorized civil engineer. We can gladly help you with respective contacts.